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Photon in a cavity – a Gedankenexperiment

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HIGHLIGHTS

- The weight of a photon in a cavity is determined in a weak gravitational field.
- Its passive gravitational mass is in agreement with Einstein's energy-mass relation.
- The cavity cannot be completely rigid.
- The interactions with the walls are controlled by momentum and energy conservation.

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ABSTRACT

The inertial and gravitational mass of electromagnetic radiation (i.e., a photon distribution) in a cavity with reflecting walls has been treated by many authors for over a century. After many contending discussions, a consensus has emerged that the mass of such a photon distribution is equal to its total energy divided by the square of the speed of light. Nevertheless, questions remain unsettled on the interaction of the photons with the walls of the box. In order to understand some of the details of this interaction, a simple case of a single photon with an energy $E_\nu = h\nu$ bouncing up and down in a static cavity with perfectly reflecting walls in a constant gravitational field $\mathbf{g}$, constant in space and time, is studied and its contribution to the weight of the box is determined as a temporal average.

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1. Introduction

Massive particles and electromagnetic radiation (photons) in a box have been considered by many authors (e.g. Poincaré, 1900; Abraham, 1904; Hasenöhrl, 1905; Einstein, 1905; Einstein, 1906; Einstein, 1908; von Laue, 1920; Bass and Schrödinger, 1955; Feenberg, 1960; Antippa, 1976; Kolbenstvedt, 1995; Rivlin, 1995; Rueda and Haisch, 2005; Ramos et al., 2011). Most of them have discussed the inertia of an empty box in comparison with a box filled with a gas or radiation.

In the presence of a constant gravitational field $\mathbf{g}$ (pointing downwards in Fig. 1), the effect on the weight of the box is another topic that has been studied. In very general terms, this problem has been treated by von Laue (1920) with the conclusion that, in a closed system in equilibrium, all types of energy E_n contribute to the mass according to

$$\Delta M = \frac{\sum E_n}{c_0^2}, \tag{1}$$

where M refers to the passive gravitational mass and c_0 is the speed of light in vacuum. For a gas and even for a single massive particle, this can easily be verified with the help of the energy and momentum conservation laws as a temporal average. For radiation, however, the situation has been debated over the years (cf., e.g. Antippa, 1976; Rivlin, 1995; Hecht, 2011).

Before we describe our thought experiment, it should be mentioned that Kolbenstvedt (1995) studied photons in a uniformly accelerated cavity and found a mass contribution

"[...] in agreement with Einstein's mass-energy formula".

This result also agrees with von Laue's equation (1) taking into account that, according to the equivalence principle (Einstein, 1911), a uniform acceleration should have the same effect on photons as a corresponding constant gravitational field. Einstein used Doppler's principle in his argumentation (cf., also Einstein, 1905). The influence of gravity could indeed be cancelled by the Doppler effect in the experiment of Pound and Rebka (1959) in such a way that the emission and absorption energies of X-ray photons generated by the 14.4 keV transition of iron (Fe^{57}) were the same in the source and the receiver positioned at different heights in the gravitational field of the Earth (Pound and Snider, 1965).

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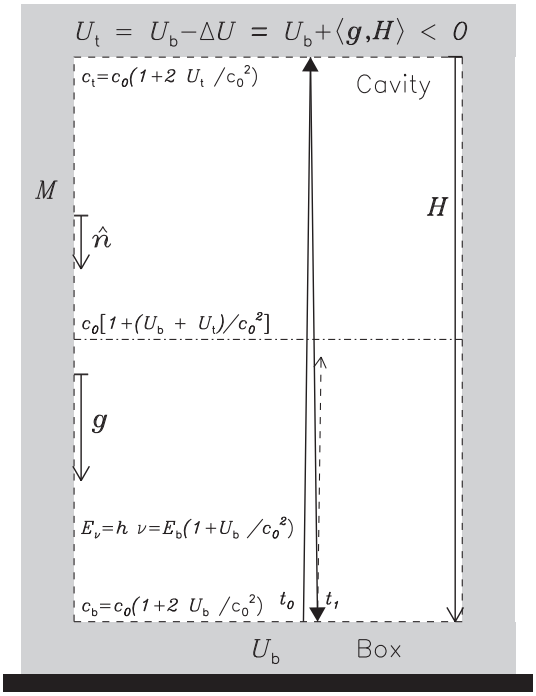


Fig. 1. Box of height $H = ||\boldsymbol{H}||$ with mass M and reflecting inner walls on a weighing scale (indicated by the black bar) to determine its weight in a constant gravitational field $\boldsymbol{g} = g\hat{\boldsymbol{n}}$. The gravitational potentials at the bottom and top are U_b and U_t, respectively. One cycle with a period of $T = t_1 - t_0$ and the continuation into the next cycle (dashed arrow) are schematically shown for a photon with an energy $E_\nu = h\,\nu$ bouncing (nearly) vertically up and down. The values of the speed of light c_b and c_t in the cavity corresponding to the potentials U_b and U_t are given at the bottom and the top as well as for the center as a mean speed of $c_0\,(1 + 2\,\overline{U}/c_0^2)$.

The experiment was described by Pound (2000) as "Weighing photons". Nevertheless, it has to be noted that the deflection angle of light passing close to the Sun deduced by Einstein (1911) with the help of the equivalence principle was only half the value he later found in his General Relativity Theory (Einstein, 1916).

It might, therefore, be instructive to consider the problem with the help of a Gedankenexperiment in the simple case of a photon bouncing up and down in a cavity of a box with perfectly reflecting inner walls *at rest* in a *constant* gravitational field. The height $\boldsymbol{H}$ is measured as fall height in the field direction, which will be indicated by the unit vector $\hat{\boldsymbol{n}}$ in equations and figures. The mass of the box, M, includes the mass of the walls and the equivalent mass of any unavoidable energy content, such as thermal radiation, except the test photon.

2. Definition of gravitational potentials

The gravitational potentials will be defined as U_t at the top of the cavity and as U_b at the bottom with

$$U_b - U_t = \Delta U = -\boldsymbol{g}\cdot\boldsymbol{H} < 0 \tag{2}$$

and the relations

$$-c_0^2 \ll U_b < \overline{U} < U_t < 0, \tag{3}$$

where c_0 is the speed of light in vacuum without a gravitational field and $\overline{U} = (U_b + U_t)/2$. Under the weak-field condition, so defined, the speed of light measured on the coordinate or world time scale is (cf. Ashby and Allan, 1979; Schiff, 1960; Guinot, 1997; Okun, 2000; Straumann, 2000)

$$c(U) = c_0\left(1 + \frac{2\,U}{c_0^2}\right). \tag{4}$$

As mentioned before, Einstein (1911) originally derived from the equivalence principle

$$c = c_0\left(1 + \frac{\phi}{c^2}\right) \tag{5}$$

with ϕ as symbol for the gravitational potential U used here. Application of Huygens' principle then led him to expect a deflection of 0.83″ for a Sun-grazing light beam. A value of 0.84″ had been obtained by Soldner (1801). However, Einstein (1916) later predicted a deflection of 1.7″ based on his general theory of relativity. This was first verified during a solar eclipse in 1919, and since then values between 1.75″ and 2.0″ have been found in many observational studies (Dyson et al., 1920; Mikhailov, 1959; Shapiro et al., 2004). Consequently, it can be concluded that observations are not consistent with Eq. (5), but are – within the uncertainty margins – in agreement with Eq. (4).

The relativistic energy (E) and momentum ($\boldsymbol{p}_0$) equation for a free body with mass m in vacuum is

$$E^2 = m^2 c_0^4 + ||\boldsymbol{p_0}||^2 c_0^2. \tag{6}$$

It reduces for a massless particle to

$$E = p_0 c_0 \tag{7}$$

with $p_0 = ||\boldsymbol{p}_0||$ (Einstein, 1917; Dirac, 1936; Okun, 1989). In a static gravitational field the energy of a photon

$$E_\nu = h\,\nu = p(U)\,c(U) \tag{8}$$

measured in the coordinate time system is constant (Okun, 2000). Its speed, however, varies according to Eq. (4), whilst the momentum changes inversely to this speed.

3. Photon reflection scenarios

At least three different scenarios can be conceived to describe the situation inside the box:

1. A naive application of Eqs. (4) and (8), i.e., direct reflections at the walls without any further interaction, leads to the result that the photon contributes

$$\Delta m \approx 2\,\frac{E_\nu}{c_0^2}\left(1 + \frac{2\,\overline{U}}{c_0^2}\right) \tag{9}$$

to the mass of the box, which is nearly a factor of two higher than expected from Eq. (1). To show this, let us consider the oppositely directed momentum transfers during photon reflections at the bottom and the ceiling of the cavity. The values are twice

$$\frac{E_\nu}{c_b}\,\hat{\boldsymbol{n}} \approx \boldsymbol{p}_\nu\left(1 - 2\,\frac{U_b}{c_0^2}\right) \tag{10}$$

and

$$-\frac{E_\nu}{c_t}\,\hat{\boldsymbol{n}} \approx -\boldsymbol{p}_\nu\left(1 - 2\,\frac{U_t}{c_0^2}\right), \tag{11}$$

where c_b and c_t are determined from Eq. (4) taking into account the conditions in (3) to justify the approximations, i.e., neglecting orders equal or higher than $(U/c_0^2)^2$ against unity. The momentum $\boldsymbol{p}_\nu = (E_\nu/c_0)\,\hat{\boldsymbol{n}}$ of a photon with energy $E_\nu = h\,\nu$ at $U_0 = 0$ in vacuum has been introduced. A complete cycle lasts for

$$T = t_1 - t_0 = \frac{2\,H}{c_0\,(1 + 2\,\overline{U}/c_0^2)}, \tag{12}$$

i.e., $2\,H$ divided by the mean speed. Within this time interval, the total momentum transfer of

$$\Delta \boldsymbol{P}^{\mathrm{C}} = 2\left(\frac{E_v}{c_\mathrm{b}} - \frac{E_v}{c_\mathrm{t}}\right)\hat{\boldsymbol{n}} \approx 4\boldsymbol{p}_v \frac{\boldsymbol{g}\cdot\boldsymbol{H}}{c_0^2}, \tag{13}$$

is obtained from Eqs. (2), (10) and (11). A momentum vector with upper index C refers in this and later equations to a momentum inside the cavity. Division by T of Eq. (12) gives a mean force of

$$\overline{\boldsymbol{F}_1} \approx 2\frac{E_v}{c_0^2}\left(1 + \frac{2\overline{U}}{c_0^2}\right)\boldsymbol{g}, \tag{14}$$

confirming Eq. (9), which has been found to be inconsistent with Eq. (1).

2. The assumption made above that the reflections occur at the walls of the box without further effects might not be correct. Consequently, the next scenario is based on an intermediate storage of the energy E_v, i.e., as elastic energy, and a transfer of a momentum calculated under the assumption that the speeds c_b and c_t valid in the cavity at U_b and U_t, respectively, are also applicable for the complete reflection process. This gives a momentum transfer of

$$2\boldsymbol{p}_v\left(1 - \frac{2U_\mathrm{b}}{c_0^2}\right) \tag{15}$$

during the absorption and emission at the bottom and

$$-2\boldsymbol{p}_v\left(1 - \frac{2U_\mathrm{t}}{c_0^2}\right) \tag{16}$$

at the top in opposite directions. Comparison with Eqs. (10) and (11) shows that the resulting force $\overline{\boldsymbol{F}_2}$ after application of Eqs. (12)–(14) equals $\overline{\boldsymbol{F}_1}$ and leads to the same surprising result.

3. The previous concept was based on the assumption that the reflections actually occur in regions of the wall, where the effective speed of light has been modified by the local gravitational potential. However, the process of a photon reflection has to be accomplished by interactions with electrons in the walls. The huge ratio of the electrostatic to the gravitational forces between elementary particles makes it unlikely that such an interaction is directly influenced significantly by weak fields of gravity. However, the *photon absorption* and *emission* will be affected as is evident from the gravitational redshift. This redshift – since its prediction by Einstein (1908) – has been theoretically and experimentally studied in many investigations (e.g. von Laue, 1920; St. John, 1928; Okun, 2000; Pound and Rebka, 1959; Schiff, 1960; Cranshaw et al., 1960; Hay et al., 1960; Blamont and Roddier, 1961; Krause and Lüders, 1961; Brault, 1963; Pound and Snider, 1965; Snider, 1972; Will, 1974; Okun et al., 1999; Wolf et al., 2010; Sinha and Samuel, 2011) resulting in a quantitative validation of Einstein's statement:

„[...] müssen also die Spektrallinien des Sonnenlichts gegenüber den entsprechenden Spektrallinien irdischer Lichtquellen etwas nach dem Rot verschoben sein, und zwar um den relativen Betrag $(\nu_0 - \nu)/\nu_0 = -\phi/c^2 = 2\cdot 10^{-6}$." (Einstein, 1911).

([...] spectral lines of light from the Sun must consequently be shifted somewhat towards the red with respect to the corresponding spectral lines of terrestrial light sources, namely by the relative value of [...] $2\cdot 10^{-6}$.)

The (negative) difference of the gravitational potentials between the Sun and the Earth is denoted by ϕ here.
Einstein's early suggestion was that the transition of an atom is an intra-atomic process, i.e. it is not dependent on the gravitational potential (Einstein, 1908):

„Da der einer Spektrallinie entsprechende Schwingungsvorgang wohl als ein intraatomischer Vorgang zu betrachten ist, dessen Frequenz durch das Ion allein bestimmt ist, so können wir ein solches Ion als eine Uhr von bestimmter Frequenzzahl ν_0 ansehen."

(Since the oscillation process corresponding to a spectral line probably can be envisioned as an intra-atomic process, the frequency of which is determined by the ion alone, we can consider such an ion as a clock with a distinct frequency ν_0.)

This means that the energy E_b initially released at the gravitational potential U_b by an elementary process equals the energy E_0 released by the *same* process at the potential $U_0 = 0$. In both cases the process will be accompanied by a momentum pair

$$\pm\boldsymbol{p}_\mathrm{b} = \pm\frac{E_\mathrm{b}}{c_0}\hat{\boldsymbol{n}} = \pm\frac{E_0}{c_0}\hat{\boldsymbol{n}}. \tag{17}$$

It has, however, to be noted in this context that later Einstein (1916) concluded:

„Die Uhr läuft also langsamer, wenn sie in der Nähe ponderabler Massen aufgestellt ist. Es folgt daraus, daß die Spektralinien von der Oberfläche großer Sterne zu uns gelangenden Lichtes nach dem roten Spektralende verschoben erscheinen müssen."

(The clock is thus delayed, if it is placed near ponderable masses. Consequently, it follows that the spectral lines of light reaching us from the surface of large stars must be shifted towards the red end of the spectrum.)

Both statements can be reconciled by postulating that the redshift occurs during the actual emission process and realizing that energy trapped in a closed system has to be treated differently from propagating radiation energy. In addition, the importance of the momentum transfer during the absorption or emission of radiation was emphasized by Einstein (1917):

„Bewirkt ein Strahlenbündel, daß ein von ihm getroffenes Molekül die Energiemenge $h\nu$ in Form von Strahlung durch einen Elementarprozeß aufnimmt oder abgibt (Einstrahlung), so wird stets der Impuls $\frac{h\nu}{c}$ auf das Molekül übertragen, und zwar bei der Energieaufnahme in der Fortpflanzungsrichtung des Bündels, bei der Energieabgabe in der entgegengesetzten Richtung. [...].
Aber im allgemeinen begnügt man sich mit der Betrachtung des Energie-Austausches, ohne den Impuls-Austausch zu berücksichtigen."

(A beam of light that induces a molecule to absorb or deliver the energy $h\nu$ as radiation by an elementary process (irradiation) will always transfer the momentum $\frac{h\nu}{c}$ to the molecule, directed in the propagation direction of the beam for energy absorption, and in the opposite direction for energy emission. [...].

However, in general one is satisfied with the consideration of the energy exchange, without taking the momentum exchange into account.)

The energy and momentum conservation principles lead to a relationship between the energy E_b and the emitted photon energy E_v at U_b of

$$E_v = E_\mathrm{b} - ||\delta\boldsymbol{p}||c_0 = ||\boldsymbol{p}_\mathrm{b} - \delta\boldsymbol{p}||c_0 = ||\boldsymbol{p}_\mathrm{b} + \delta\boldsymbol{p}_\mathrm{b}||c_\mathrm{b} = ||\boldsymbol{p}(U_\mathrm{b})||c_\mathrm{b} \tag{18}$$

by introducing a differential momentum $\delta\boldsymbol{p}_\mathrm{b}$ parallel to $\boldsymbol{p}_\mathrm{b}$ (neglecting the very small recoil energy). The evaluation gives, together with Eq. (4),

$$\delta\boldsymbol{p}_\mathrm{b} = -\boldsymbol{p}_\mathrm{b}\frac{U_\mathrm{b}}{c_0^2} \tag{19}$$

and

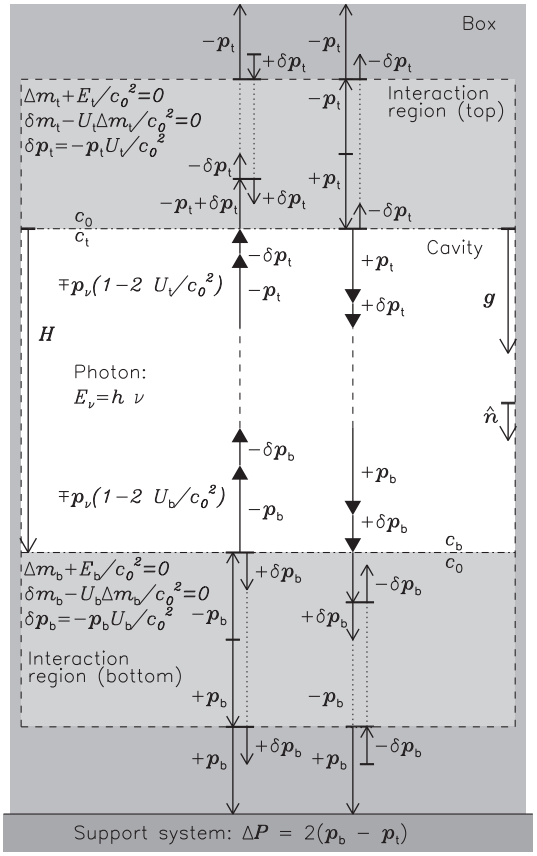


Fig. 2. The box is shown on a weighing scale as support system. The bottom and the top of the cavity are complemented by "interaction regions". Energy release and photon emission processes occur in these regions as required by *energy and momentum conservation*. The initial energy release of $E_b = \Delta m_b c_0^2$ happens in the bottom interaction region accompanied by a momentum transfer to the box and the support system of $+\boldsymbol{p}_b$. The conversion of Δm_b into energy at U_b is subject to an increase of the potential energy of the bottom interaction region by $-U_b \Delta m_b$ taken from E_b. The related differential momentum contribution $+\delta\boldsymbol{p}_b$ acts on the support system. It will be compensated by the opposite effect of the following energy-to-mass conversion at U_b after one period T. The corresponding steps at U_t are described in the text.

$$E_\nu = E_b\left(1 + \frac{U_b}{c_0^2}\right). \tag{20}$$

Such a scenario has recently been discussed in the context of the gravitational redshift (Wilhelm and Dwivedi, 2014).
With the basic assumption that the photon is reflected from both walls with the same energy E_ν, the relation for the ceiling is:

$$E_\nu = E_t\left(1 + \frac{U_t}{c_0^2}\right). \tag{21}$$

The corresponding momentum transfers during absorption and emission expected according to Eqs. (17), (20) and (21) are

$$2\boldsymbol{p}_b \approx 2\frac{E_\nu}{c_0}\left(1 - \frac{U_b}{c_0^2}\right)\hat{\boldsymbol{n}} = 2\boldsymbol{p}_\nu\left(1 - \frac{U_b}{c_0^2}\right) \tag{22}$$

at the bottom and

$$-2\boldsymbol{p}_t \approx -2\frac{E_\nu}{c_0}\left(1 - \frac{U_t}{c_0^2}\right)\hat{\boldsymbol{n}} = -2\boldsymbol{p}_\nu\left(1 - \frac{U_t}{c_0^2}\right) \tag{23}$$

at the top. Note in this context that the elementary processes at the bottom and the top must have slightly different energy levels for constant E_ν. Comparison of the momentum values of Eqs. (22) and (23) with those of Eqs. (10) and (11) shows that the force $\overline{\boldsymbol{F}_3}$ obtained in analogy to Eq. (13) equals $\overline{\boldsymbol{F}_1}/2$ and thus gives a value expected from Eq. (1).
This encouraging result will now be analysed in detail. The initial energy release is assumed in the "Interaction region (Bottom)" at the potential U_b in Fig. 2 on the left according to

$$E_b + \Delta m_b c_0^2 = 0 \tag{24}$$

accompanied by a momentum pair of $\pm\boldsymbol{p}_b$ as mentioned above. The exact release process is of no importance for the present discussion, but the initial release must be controlled by the elementary process alone. A speed of c_b at the bottom of the cavity, together with energy and momentum conservation laws, then requires that a photon can only be emitted with an energy E_ν given by Eqs. (18) and (20) (the rest energy of the mass Δm_b at the gravitational potential U_b) and a momentum in the upward direction of

$$-\boldsymbol{p}_b^C = -\boldsymbol{p}_b - \delta\boldsymbol{p}_b \approx -\boldsymbol{p}_\nu\left(1 - \frac{2U_b}{c_0^2}\right), \tag{25}$$

where we have used Eqs. (19) and (22). The energy difference, corresponding to the potential energy of a mass Δm_b at U_0 relative to U_b,

$$E_b - E_\nu = -U_b \Delta m_b \tag{26}$$

will be transferred to the box. This process is accompanied by a corresponding differential momentum transfer of $+\delta\boldsymbol{p}_b$ and by a mass increase of

$$\delta m_b = -\frac{U_b \Delta m_b}{c_0^2}. \tag{27}$$

The momentum $+\boldsymbol{p}_b$ from the initial release and $+\delta\boldsymbol{p}_b$ will thus be acting on the support system.

The reflections at the top and bottom have to be considered in several steps – illustrated in detail in Fig. 2:

- The photon E_ν arrives at the top with c_t, obtained from Eq. (4), and a momentum of

$$-\boldsymbol{p}_t^C = -\boldsymbol{p}_b\frac{(1 - U_b/c_0^2)(1 + 2U_b/c_0^2)}{1 + 2U_t/c_0^2} \approx -\boldsymbol{p}_t - \delta\boldsymbol{p}_t \approx -\boldsymbol{p}_\nu\left(1 - \frac{2U_t}{c_0^2}\right) \tag{28}$$

derived from Eqs. (8) and (23) with

$$\delta\boldsymbol{p}_t = -\boldsymbol{p}_t\frac{U_t}{c_0^2}. \tag{29}$$

- The change of the speed from c_t to c_0 in the interaction region will entail a change of the momentum in Eq. (28) to a new value, which can be written in our approximation as $-\boldsymbol{p}_t + \delta\boldsymbol{p}_t$.
- The energy-to-mass conversion at the potential U_t according to

$$\Delta m_t + \frac{E_t}{c_0^2} = 0 \tag{30}$$

in the upper interaction region can only be accomplished by adding the potential energy term

$$E_t - E_\nu = -U_t \Delta m_t \tag{31}$$

and a momentum of $-\delta\boldsymbol{p}_t$, which will be provided by the conversion of δm_t into energy. The momentum $+\delta p_t$ of the momentum pair will act on the box. In total a momentum of $-\boldsymbol{p}_t + \delta\boldsymbol{p}_t$ has thus to be taken up by the box.
These processes can be seen as the reversed actions performed by Eqs. (24)–(27), but now at U_t.

- The return trip essentially occurs in the reverse order as shown on the right side of Fig. 2. The energy release of E_t will be accompanied by a momentum pair $\pm\boldsymbol{p}_t$. The photon will be emitted in the downward direction with a momentum of

$$\boldsymbol{p}_t^C = +\boldsymbol{p}_t + \delta\boldsymbol{p}_t \approx +\boldsymbol{p}_v\left(1 - \frac{2U_t}{c_0^2}\right), \tag{32}$$

cf., Eqs. (23) and (28). The energy difference $E_t - E_v$ restores the potential energy of Eq. (31). A momentum of $-\boldsymbol{p}_t - \delta\boldsymbol{p}_t$ will be transferred to the box in analogy to the processes at U_b.
- The photon momentum at the bottom of the cavity will be

$$\boldsymbol{p}_b^C = \boldsymbol{p}_t\frac{(1 - U_t/c_0^2)(1 + 2U_t/c_0^2)}{1 + 2U_b/c_0^2} \approx +\boldsymbol{p}_b + \delta\boldsymbol{p}_b$$
$$\approx +\boldsymbol{p}_v\left(1 - \frac{2U_b}{c_0^2}\right) \tag{33}$$

as expected from Eq. (25).
- In analogy to the situation at U_t, we find a momentum of $+\boldsymbol{p}_p - \delta\boldsymbol{p}_b$ that has to be transferred to the support system.
- The total external momentum thus is with Eqs. (2), (3), (17), (20) and (21):

$$\Delta\boldsymbol{P} = (\boldsymbol{p}_b + \delta\boldsymbol{p}_b) - (\boldsymbol{p}_t - \delta\boldsymbol{p}_t) - (\boldsymbol{p}_t + \delta\boldsymbol{p}_t) + (\boldsymbol{p}_b - \delta\boldsymbol{p}_b)$$
$$= 2(\boldsymbol{p}_b - \boldsymbol{p}_t) = 2\left(\frac{E_b}{c_0} - \frac{E_t}{c_0}\right) \approx -2\frac{E_v}{c_0}\frac{\Delta U}{1 + 2\overline{U}/c_0^2}$$
$$= 2\frac{E_v}{c_0}\frac{\boldsymbol{g}\cdot\boldsymbol{h}}{1 + 2\overline{U}/c_0^2}. \tag{34}$$

Averaged over the time interval T from Eq. (12) for a full cycle gives a mean force of

$$\overline{\boldsymbol{F}_3} = \frac{\Delta\boldsymbol{P}}{T} \approx \frac{E_v}{c_0^2}\boldsymbol{g} \tag{35}$$

in agreement with Eq. (1).
A short comment is required on the intermediate mass storage processes at the bottom and the top. Compared to T the storage times are so small that the contributions to the mean mass and thus the weight can be neglected. Since a factor of two was in question, there was also no need to complicate the calculations even further by retaining terms which are very small under the weak-field conditions assumed.

A multi-step process including Einstein's assumption of an intra-atomic energy liberation thus leads to the correct result within our approximations. It involves an external box which cannot be completely rigid as it must be able to enact the required gravitational energy and momentum transfers. Another objection against a rigid box is the limited speed of any signal transmission in the side walls (Antippa, 1976; Hecht, 2011).

4. Discussion

The assumption of a constant field $\boldsymbol{g}$ in the environment of the box – made in the interest of simplifying the calculations – calls for some explanations. Einstein (1912) discussed a static gravitational field without mass („[...] ein massenfreies statisches Gravitationsfeld [...]") and clarified in a footnote that such a field can be thought of as generated by masses at infinity. Bondi (1986) went even further in his scepticism against constant gravitational fields. *Only* the non-uniformity of the field would be observable. A field with constant magnitude and direction should not be regarded as a field.

Indeed, there can be no perfectly homogeneous gravitational field. Nevertheless very good approximations remote from the generating masses exist between neighbouring potential surfaces, and the effects *are* observable. These remarks are of relevance for the considerations in the previous section, as the calculations could have significantly been shortened by setting $U_t = 0$, but no realistic configuration would correspond to such a definition. It can thus be concluded that Eq. (2) should be amended by $|\Delta U| \ll |U_b| \approx |U_t|$ in order to justify the assumption of a (nearly) constant field.

We (Wilhelm and Dwivedi, 2014) argued that the gravitational redshift has some resemblance with the Doppler effect as discussed by Fermi (1932), if the interaction of the liberated energy during an atomic transition with the *kinetic energy* of the emitter and its momentum is replaced by the *potential energy*. The same argument appears to be relevant for the photon reflection process discussed here.

5. Conclusion

The temporally averaged increase of the passive gravitational mass and thus the weight of a box containing a single photon bouncing up and down in a weak gravitational field could be obtained in agreement with Eq. (1) by assuming interactions between the photon and the walls of the cavity as well as momentum and energy conservation.

Preston (1883) formulated and wished some 130 years ago:

"[...] let us work towards the great generalization of the Unity of Matter and Energy."

Most of the work has been done by now, but it has also become quite clear that the interactions between photons and matter in a gravitational field are complicated as outlined by Bondi (1997) in his article entitled

"Why gravitation is not simple."

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